Aggregate Fluctuations and Labor Market Adjustment

Labor II — Week 10

Teaching deck draft

Central question and week repositioning

- When aggregate conditions weaken or strengthen, which labor-market margins do the adjusting?
- Week 10 scales Week 4 search objects and Week 9 institutional margins up to economy-wide shocks.
- The lecture stays labor-focused: unemployment, vacancies, separations, job-to-job mobility, wages, and match quality.

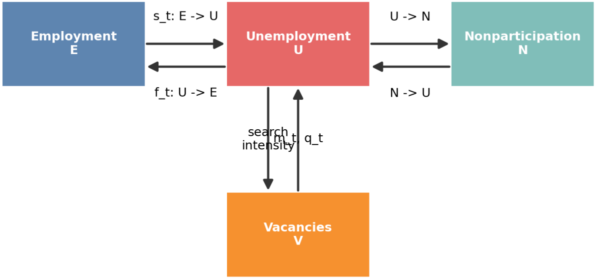
Week 9 to Week 10 bridge

- Week 9: regulation, enforcement, and insurance shape layoffs, recalls, search, and vacancy posting.
- Week 10: the same margins move when shocks are aggregate rather than policy-specific.
- Week 11 will keep these margins in view but switch from cyclical adjustment to structural technology shocks.

Stocks, flows, and hidden margins

Stocks

U, E, N, and V are visible states.



Hazards and hidden margins

Job finding, separation, participation, and job-to-job mobility determine how the stock moves.

Labor-market stocks and the law of motion for unemployment

$$\dot{u}_t = s_t(1 - u_t) - f_t u_t, \qquad u_t^* = \frac{s_t}{s_t + f_t}$$

- Unemployment is a stock.
- Job finding f_t and separation s_t are hazards.
- A high unemployment rate can come from high inflows, low outflows, or both.

Three-state extension and hidden slack

$$\begin{pmatrix} E_{t+1} \\ U_{t+1} \\ N_{t+1} \end{pmatrix} = P_t \begin{pmatrix} E_t \\ U_t \\ N_t \end{pmatrix}$$

- The $E/U/N$ system keeps participation margins visible.
- A pure two-state unemployment narrative can miss hidden slack and attachment dynamics.
- Gross worker flows matter for aggregate labor adjustment, not only the unemployment rate.

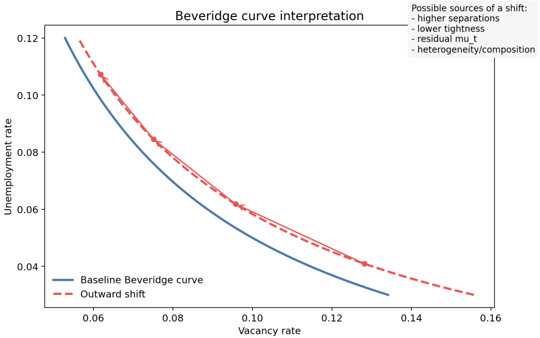
Matching, tightness, and the Beveridge curve

$$m_t = \mu_t u_t^{\alpha} v_t^{1-\alpha}, \qquad f_t = \frac{m_t}{u_t} = \mu_t \theta_t^{1-\alpha}, \qquad q_t = \frac{m_t}{v_t} = \mu_t \theta_t^{-\alpha}$$

- Tightness $\theta_t = v_t / u_t$ links worker-side and vacancy-side hazards.
- Workers care about f_t; firms care about q_t.
- Beveridge-curve shifts need not identify one primitive.

Matching residuals and Beveridge-curve interpretation

$$\mu_t = \frac{m_t}{u_t^{\alpha} v_t^{1-\alpha}}, \qquad u_t^* = \frac{s_t}{s_t + \mu_t \theta_t^{1-\alpha}}$$



Free entry, surplus, and the Shimer-Hall logic

$$\kappa = q_t \, \mathbb{E}_t[J_{t+1}], \qquad S_t = J_t + W_t - U_t$$

- Vacancy creation depends on expected surplus.
- Shimer: benchmark shocks do not move surplus enough to match vacancy and unemployment volatility.
- Hall: wage stickiness keeps wages from absorbing the whole shock and amplifies vacancy posting.

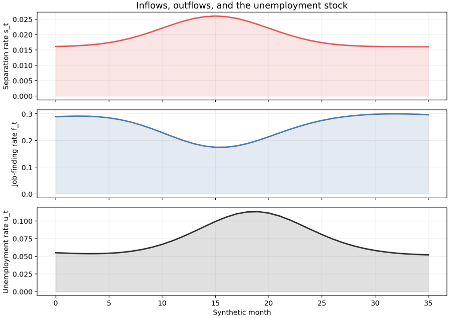
Unemployment stocks versus flows

$$\Delta u_t \approx (1 - u_t)\Delta s_t - u_t \Delta f_t$$

- Elsby-Michaels-Solon: cyclical unemployment is an inflow-outflow object.
- Severe recessions often involve both weaker job finding and higher separations.
- CPS-style decompositions observe transition hazards, then infer contributions to the stock.

Separations, endogenous destruction, and churning

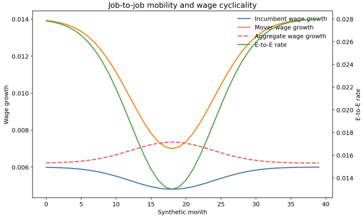
$$s_t = \Pr(S_t(z) < 0)\,, \qquad \text{Churning}_t = \text{Worker Reallocation}_t - \text{Job Reallocation}_t$$



On-the-job search, job-to-job transitions, and churning

$$EE_t = \lambda_t^e \Pr\big(w' > w\big)$$

- Aggregate labor adjustment is not only unemployment.
- Job ladders flatten in weak markets when outside offers collapse.
- Reallocation among employed workers is a wage and match-quality margin.

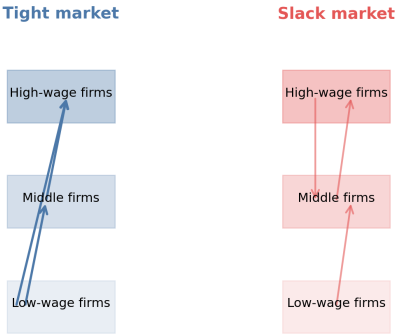


Wage cyclicality, sorting, and match quality

$$\Delta \log \bar{w}_t = \beta^I \, cyc_t + \omega_t^{EE} \beta^{EE} \, cyc_t + comp_t + sort_t$$

- Incumbent wage cyclicality is not the same object as mover wage cyclicality.
- Average wage growth is contaminated by selection and sorting.
- Weak measured wage cyclicality may hide strong movements in outside offers and match quality.

Match quality and cyclical reallocation



Tight markets: more upgrading, more poaching, stronger wage ladders.

Slack markets: weaker outside offers, less upgrading, more composition bias.

Empirical designs and what they identify

Design	Unit	Observed margin	Main hidden margin
National time series	market-time cell	U, V, hires, flows	composition and segmentation
Stock-flow decomposition	worker-state transitions	$U \to E$, $E \to U$, U stock	recall, misclassification, job quality
State/local cyclicality	area-time cell	unemployment, wages, vacancies	aggregate equilibrium mechanism
Matched employer-employee data	worker-firm spell	$E \to E$, separations, wage ladder	full opportunity set and GE response

Recent Beveridge-curve-shift evidence and open questions

- A matching residual is not enough: heterogeneity, search intensity, and vacancy composition all matter.
- Why do Beveridge-curve shifts differ across eras?
- How much cyclical adjustment runs through recalls, temporary layoffs, and participation margins?
- How much measured wage rigidity is really sorting or match-quality change?

Bridge to Week 11 technology and adjustment

- Week 10: cyclical shocks move hazards, vacancies, mobility, and wages over the business cycle.
- Week 11: technology, automation, and AI move many of the same labor-market objects through more persistent restructuring.
- The labor-market margins stay the same even when the shock changes.

Takeaways

- Unemployment is a stock governed by multiple hazards.
- Matching, separation, and job-to-job mobility are distinct cyclical margins.
- Beveridge-curve shifts and wage cyclicality are hard to interpret without composition and sorting.
- Aggregate labor adjustment is a map of labor-market objects, not a single macro fact.

Appendix: Week 10 lab bridge

- Reproduce: a Barnichon-Figura-style aggregate matching exercise.
- Diagnose: an Elsby-Michaels-Solon-style unemployment stock-flow decomposition.
- Transfer: a small job-to-job and wage-cyclicality exercise.
- Keep the object explicit: stock, hazard, efficiency residual, or job ladder.